

Vision, Scope, and Plan Document

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Table of Contents

Problem Statement.....	3
Project Background.....	3
Stakeholders.....	3
Users	3
Risks.....	3
Assumptions.....	3,4
Vision of the Solution.....	4
Vision Statement.....	4
List of Features.....	4
Desirable Features.....	4
Project Plan	5
Statement of Work (SOW).....	5
Resource List	5
Work Breakdown Structure (WBS)	6
Schedule.....	7
Risk Plan	7,8

Problem Statement

Project Background

The purpose of our project is to create a children's toy designed to introduce young children to the letters of the alphabet, while providing the option of games that will encourage critical thinking and allow for the enhancement of vocabulary. We hope to take advantage of technology to better interest and engage young children in learning.

Stakeholders

- Andy Nguyen: Project Leader/Hardware Team
- John Baumann: Hardware Team
- Sean Lovstuen: Software Team
- Syed Nayeem: Software Team
- Khoi Pham: Software Team

Users

Children between the ages of 2-5

Risks

- Covid-19
- Hardware risks
 - Board being wired incorrectly
 - Nonfunctioning hardware parts
 - Hardware incompatibility
- Software risks
 - Coding/Debugging complications
 - Incompatibility between software and hardware
 - IDE may be inadequate for project
- Design risks
 - Toy breaking apart
 - Toy body is unstable
 - Dismemberment of toy
 - Accidental dislodging of toy pieces

Assumptions

- Required materials for project will be obtained
- Hardware components are in working condition
- Any face-to-face meetings will be done with precaution and follow Covid-19 safety guidelines
- Team members may have personal conflicts that could affect the project

- Team member understands the current situation and will be lenient about any unforeseen circumstance(s)
- Team members understand their role within the team
- Team meeting place will be provided

Vision of the Solution

Vision Statement

Our vision is to create a world where everyone has equal opportunity and access to affordable, high-quality, education. In today's society, the means of a higher education comes with a high cost. Being able to afford a degree means having someone that is able to pay for the education, or taking on a ridiculous amount of debt in hopes of being able to hold a piece of paper that validates their worthiness.

Our goal is to eliminate the norm of what it means to be truly educated. Subjects like math and science are important to learn but skills like empathy and compassion are ones that are hardly emphasized in schools. Education is never-ending and goes far beyond the walls of a classroom door and that's where we come in. We wish to close the gap on the unbalance of access to education based on income and to provide more than what a typical education system has to offer.

We wish to tackle this problem by providing education through unique and interactive toys. Our vision will take the meaning of "toy" to a new and unknown level. Developing toys that have the possibility of creating everlasting impact on a person's life. Toys that will provide and encourage mental growth and critical thinking. The future holds a world where our products will range from teaching young children the alphabet to teaching adults about deducting taxes, and everything in-between.

List of Features

- Learning Mode
- Quiz Mode
- Holographic display
- Rotating case
- 26 alphabet buttons
- 2 mode changing buttons
- Power button

Desirable Features

- More games for Quiz Mode (Hangman, spelling bee, etc.)
- Additional screens
- Additional buttons

Project Plan

Statement of Work (SOW)

Deliverable Description	Criteria	Project Worker(s)
Gather materials	- Must be Arduino compatible - Must be affordable	Any Team Member
Arduino code to switch between buttons	- The letters must correspond with the first letter of a presented word	Software Team
Arduino code to display image	- Image must fit within the display screen	Software Team
Arduino code to play audio	- Audio must be loud enough to be heard at a reasonable distance	Software Team
Display of a holographic image	- Image must be in the proper format for hologram	Software Team
Arduino code for Quiz “A”	- Must be a game that displays an image for the user to choose the correct starting letter of the image displayed	Software Team
Arduino board to display	- Board must be properly connected - Risk safety must be taken into account	Hardware Team
Arduino board to play audio	- Board must be properly connected - Risk safety must be taken into account	Hardware Team
Connecting Arduino board to all hardware parts	- All parts must be in their proper place and properly connected	Hardware Team
Sizing of plastic case for hologram image	- Case must be big enough to house all components - Opening must be big enough display the entire image	Hardware Team

Resource List

- Arduino Hardware
- Arduino compatible LCD
- Arduino compatible speaker
- microSD card
- PC with Arduino SDK
- 3D printer for building toy case
- Clear plastic for holographic project

Work Breakdown Structure (WBS)

1. Gather materials - 1 hour(s)
 - 1.1 Arduino Mega 2560 Rev 3 (~\$40.00)
 - 1.2 5" TFT LCD Display (~\$40.00 - ~\$90.00)
 - 1.3 Buttons (~\$6.00 - \$13.00)
 - 1.4 Mini Breadboards (~\$10.00 for 12 pack)
 - 1.5 22 AWG stranded wires (~\$5.00 - \$10.00)
 - 1.6 Speakers (~\$6.00 - \$10.00)
2. Arduino code to switch between buttons - 5-6 hour(s)
 - 1.1 Research connection between board and button
 - 1.2 Connect board to a single button
 - 1.3 Create a switch case for each button
 - 1.4 Connect board to multiple buttons
3. Arduino code to display image - 4-5 hour(s)
 - 1.1 Research connection between board and display
 - 1.2 Connect board to display screen
 - 1.3 Create a function that will display image depending on button
 - 1.4 Gather all appropriate images needed
4. Arduino code to play audio - 4-5 hour(s)
 - 1.1 Research connection between board and audio device
 - 1.2 Connect speaker to board
 - 1.3 Create a function that will display specific audio depending on button
 - 1.4 Gather all appropriate audio needed
5. Display of a holographic image - 8 hour(s)
 - 1.1 Research clear plastic size depending on display screen size
 - 1.2 Research how to create image to display
 - 1.3 Research ways to save images to board
6. Arduino code for Quiz "A" - 12 hour(s)
 - 1.1 How to get to Quiz mode
 - 1.2 What will happen during quiz game
7. Arduino board to display - 3 hour(s)
 - 1.1 Connect display to board
8. Arduino board to play audio - 3 hour(s)
 - 1.1 Connect speaker to board
9. Connecting Arduino board to all hardware parts - 3 hour(s)
 - 1.1 Soldering the loose parts to the board
 - 1.2 Connecting other hardware to the board
10. Sizing of plastic case for hologram image - 5 hour(s)
 - 1.1 Calculating the size of the screen to figure out the dimensions of the clear plastic case
 - 1.2 Cutting the plastic to shape
 - 1.3 Attach the case to the screen

Schedule

- Gather materials
- Create all physical buttons
- Assemble all Arduino parts
- Properly display a holographic image
- Develop algorithm for Learning Mode
 - Create a database of letters with their relevant attributes (audio pronunciation, holographic image)
 - Get each physical button to correspond with a letter object in the database
 - Display chosen holographic letter
 - Output sound pronouncing chosen letter
- Develop algorithm for switching modes
- Develop algorithm for Quiz Mode
 - Create database of quiz words with their relevant attributes (difficulty, audio, holographic image)
 - Create difficulty/subject selection screen
 - Create code to select a certain number of words from the database to make up the quiz
 - Test quiz
 - Create audiovisual feedback to inform the child that they gave a correct/incorrect answer
- Complete product model design
- Fully assemble toy

Risk Plan

Risk	Description	Solution(s)
COVID-19	Team Meetings	- Members will gather only if necessary - Members will abide strictly by Covid-19 guidelines - Members will not attend meetings if he/she believes they have been exposed to Covid-19 or exposed to someone they believe may have it - Members will not attend meetings if he/she believes they are currently exhibiting symptoms - A member fallen to any sort of Covid-19 exposure or symptoms MUST notify all Team Members immediately
Hardware	Improper hardware handling	- Hardware members should review the studies of digital circuits - Hardware members should be fully prepared before trying to implement a design - Hardware members should communicate any developing problems as clear and concise as possible
Hardware	Nonfunctional pieces	- Although it is assumed that hardware parts are functional, hardware members should always test each piece to check for defects

		- If a nonfunctional hardware piece is found, it should be removed from the project immediately and be disposed/recycled of properly
Hardware	Incompatibilities	- Raspberry Pi will be considered as an alternative if Arduino board is incapable of handling project stress - Incompatibility with other hardware from a change must also be taken into consideration - There are a variety of computing hardware to choose from, minimizing the risk of hardware incompatibilities
Software	Code development problems	- Software members should review the studies of Arduino IDE - Software members should consistently be testing product code throughout the software development lifecycle - Software members must be able to coordinate sharing, editing, and testing different code for a single Arduino board - Software members should communicate any developing problems as clear and concise as possible
Design	Childproof	- Hardware members must take the time to refine internal body of toy for all possible safety concerns - Hardware members must make the holographic project of the product both child-safe and tamper-proof - Hardware members must make the body of the toy durable enough to handle stress but not heavy enough to be dangerous - Software members must take the time to iterate through developed code and check for any incompatibilities with hardware devices that may be of safety concerns.